

health
Department:
Health
REPUBLIC OF SOUTH AFRICA

Spartcon
TECHNOLOGIES

MIDDELBURG PHARMACEUTICAL FACILITY

BMS Dashboard User Guide

How to monitor the warehouse and operate the HVAC units safely

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Prepared by Spartcon Technologies for the Department of Health · Middelburg

Handover edition · 11 November 2026

1 Getting started

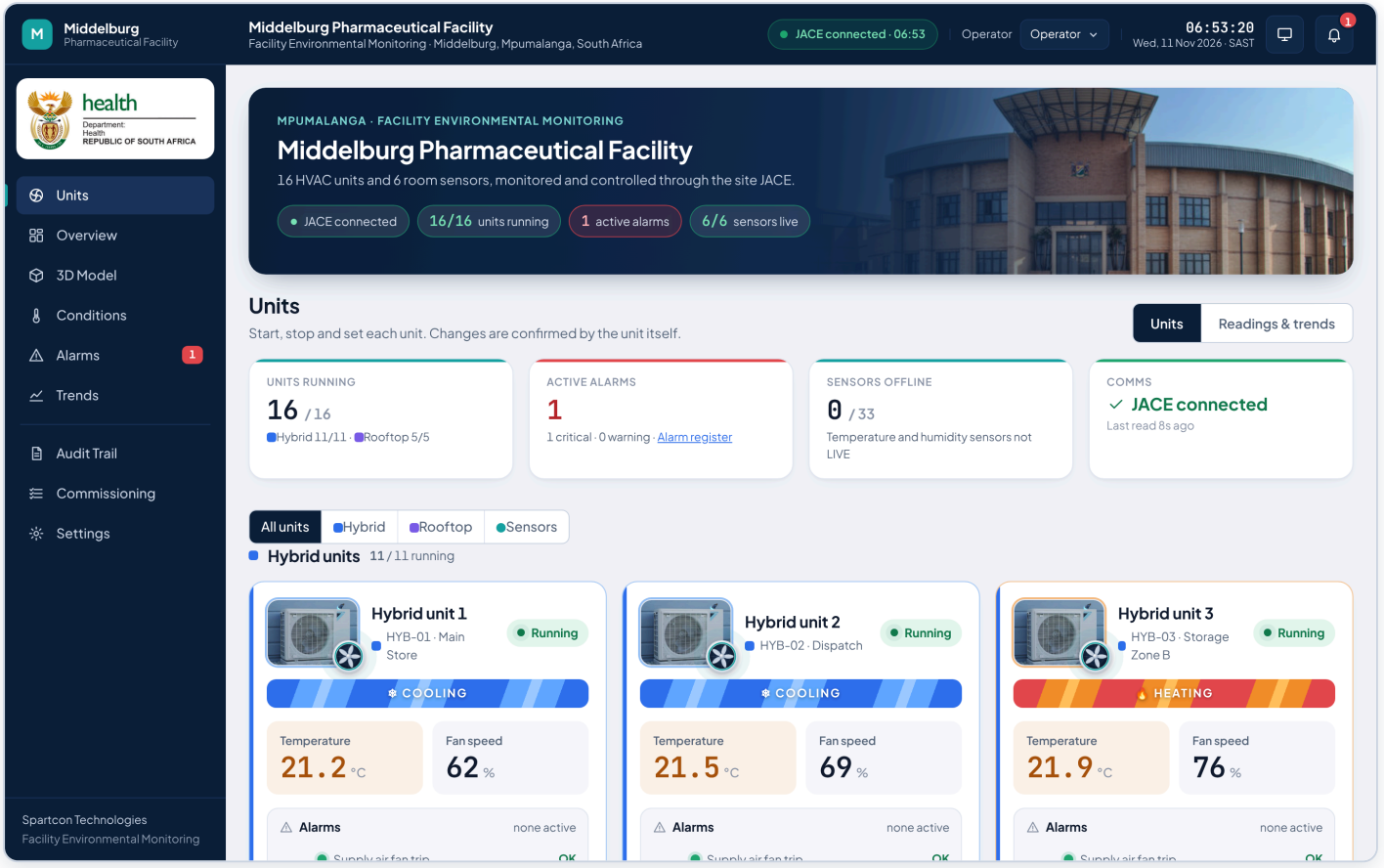
The BMS dashboard shows the live temperature, humidity and status of every HVAC unit and room sensor in the warehouse, records every reading, and lets authorised operators start/stop the rooftop units and adjust their temperature setpoints.

Open the dashboard

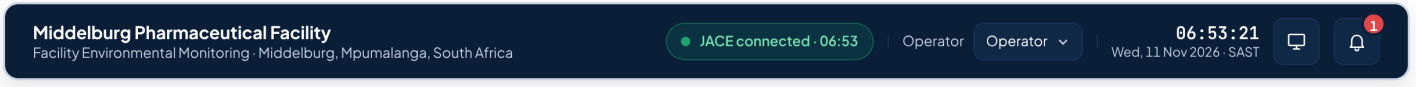
- **On site** (any PC on the site network): **http://192.168.1.90:8080**
- **Remotely**: connect Tailscale first (see the Remote Access guide), then **http://100.103.134.56:8080**
- Use Chrome, Edge, Firefox or Safari. No software to install.

Who can do what

- **Viewer**: sees everything, cannot send commands.
- **Operator**: can start/stop rooftop units and change setpoints.
- **Admin**: Spartcon only (settings).



The Units page (home). Left: menu. Top: connection status, your role, clock, wall-display mode and the alarm bell.



Green "JACE connected"

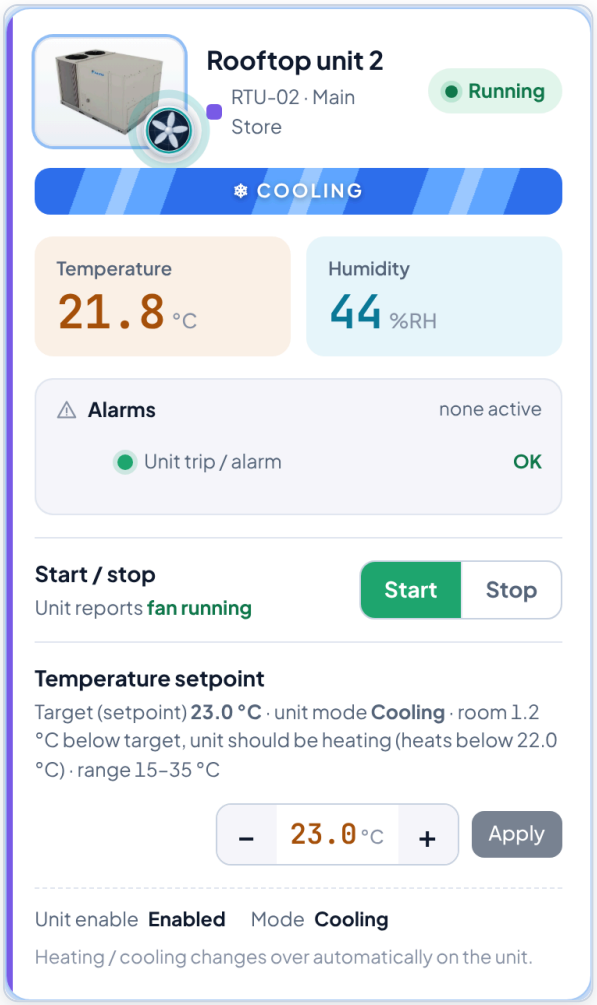
Readings are live from the building controller. The clock shows when it last read.

Red "JACE connection lost"

Values are no longer live and controls are locked. Contact the IT administrator.

2 The Units page

Every HVAC unit has its own card. The tiles at the top summarise the whole site: units running, active alarms, sensors offline and the connection.



Reading a unit card

1. **Name and zone**, e.g. Rooftop unit 2 · Main Store.
2. **State**: Running · Stopped · Fault · No data
3. **Fan badge** on the photo spins only while the unit's fan is running.
4. **Mode band**: blue COOLING or amber HEATING while the unit runs in that mode.
5. **Temperature and humidity** measured by the unit.
6. **Alarms**: each alarm point, green OK or red ALARM (page 7).
7. **Start / stop** buttons (page 4).
8. **Temperature setpoint** with − / + and Apply (page 5).
9. **Facts**: unit enable and the mode reported by the unit.

Filter the list

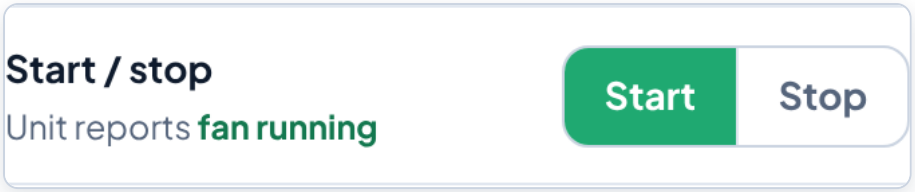
Use **All units / Hybrid / Rooftop / Sensors** above the cards. Click a unit's **name** for its detail panel with a 1-hour temperature trend.

A dash "—" means no live value

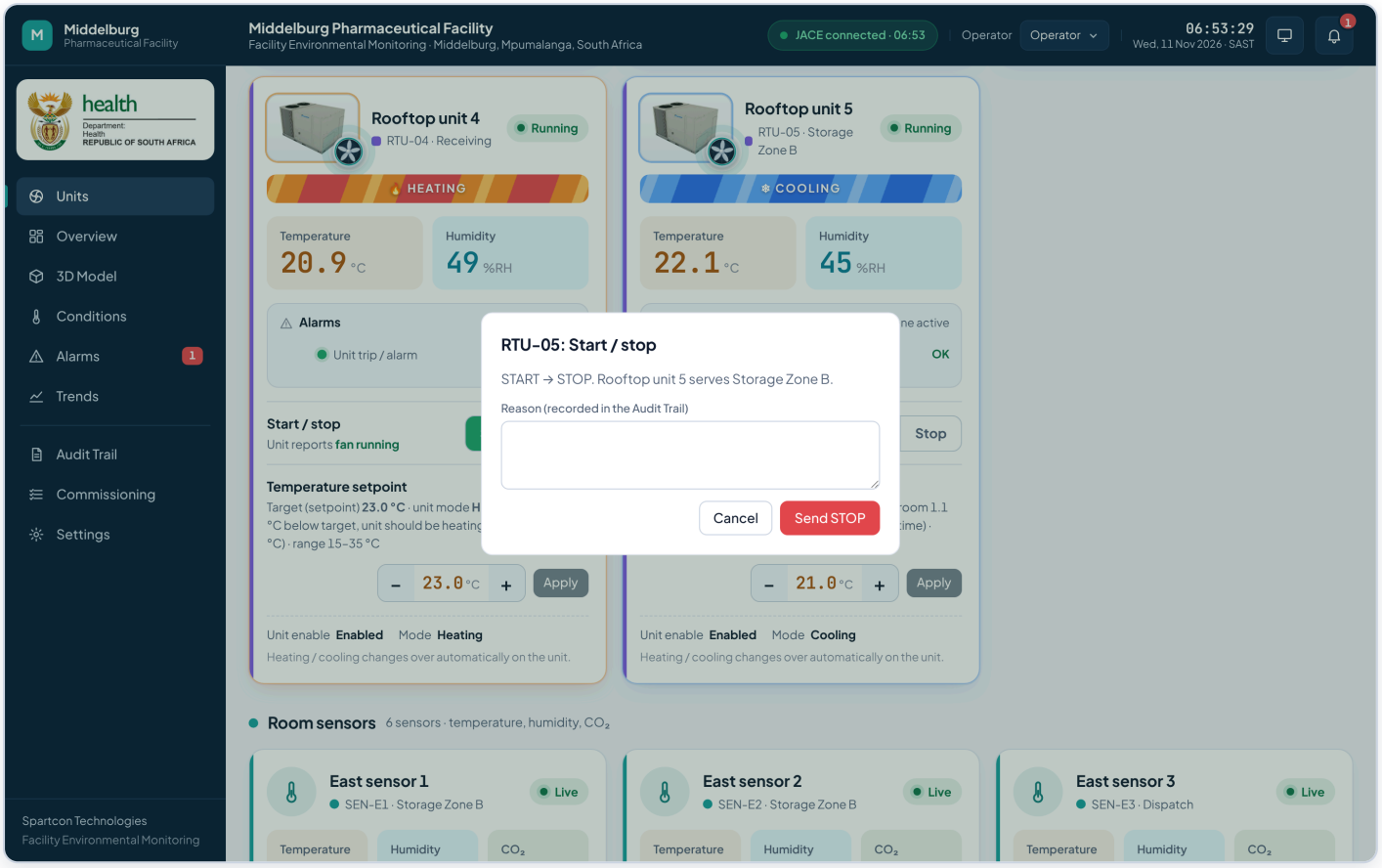
The grey label beside it says why (STALE, OFFLINE…). The dashboard never shows an old value as current.

3 Start or stop a rooftop unit

Only the 5 rooftop units can be started or stopped from the dashboard. You must be logged in as **Operator**.



1. Find the unit on the **Units** page (filter **Rooftop**).
2. Click **Stop** (or **Start**). The highlighted button shows the current command.
3. A confirmation window opens. Check the unit and the change, add a reason if you like (it is saved in the Audit Trail), then click **Send**. Click **Cancel** if you are unsure.
4. Watch the message under the button until it turns green.



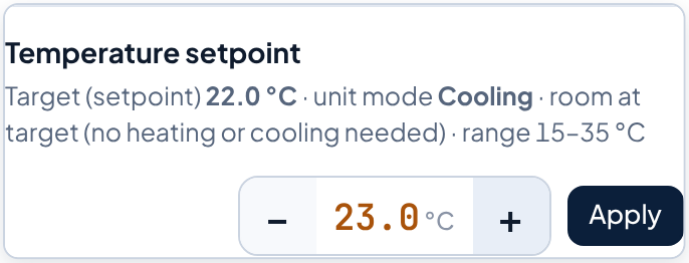
Every command asks for confirmation first.

Message	Meaning
Sending…	Sent to the controller, waiting for it to confirm (normally a few seconds).
Confirmed	The controller accepted the command.
Command accepted, unit not following	The controller has the command but the unit's fan has not followed. Check the unit / its local controls and call the IT administrator.
Not confirmed / Failed / Not sent	The command did not take effect. The message says why. Do not keep pressing the button; call the IT administrator.

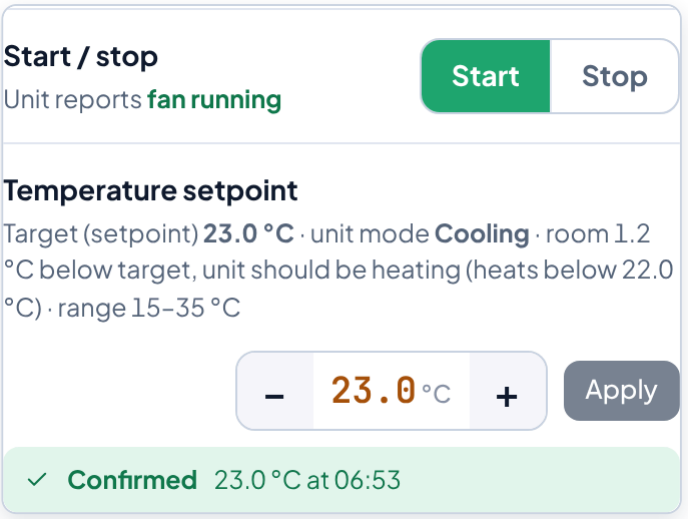
4 Change the temperature setpoint

The setpoint is the **target temperature** for a rooftop unit (15–35 °C, steps of 0.5 °C). The unit then heats or cools by itself to reach it.

1. On the rooftop unit's card, press **−** or **+** until the new target shows.
2. Click **Apply**.
3. Check the change in the confirmation window (old → new) and click **Send**.
4. Wait for **Confirmed** under the setpoint.



Step 1: the new target (23.0 °C) is chosen but not sent yet.



Step 4: the controller confirmed the new setpoint.

How heating and cooling work (automatic)

- Room **above** the setpoint → the unit **cools**.
- Room more than **1 °C below** the setpoint → the unit **heats**.
- In between → neither (this stops it switching back and forth).

Example: room 25 °C, setpoint 18 °C → cooling. The card tells you which way the unit should be working.

Be patient

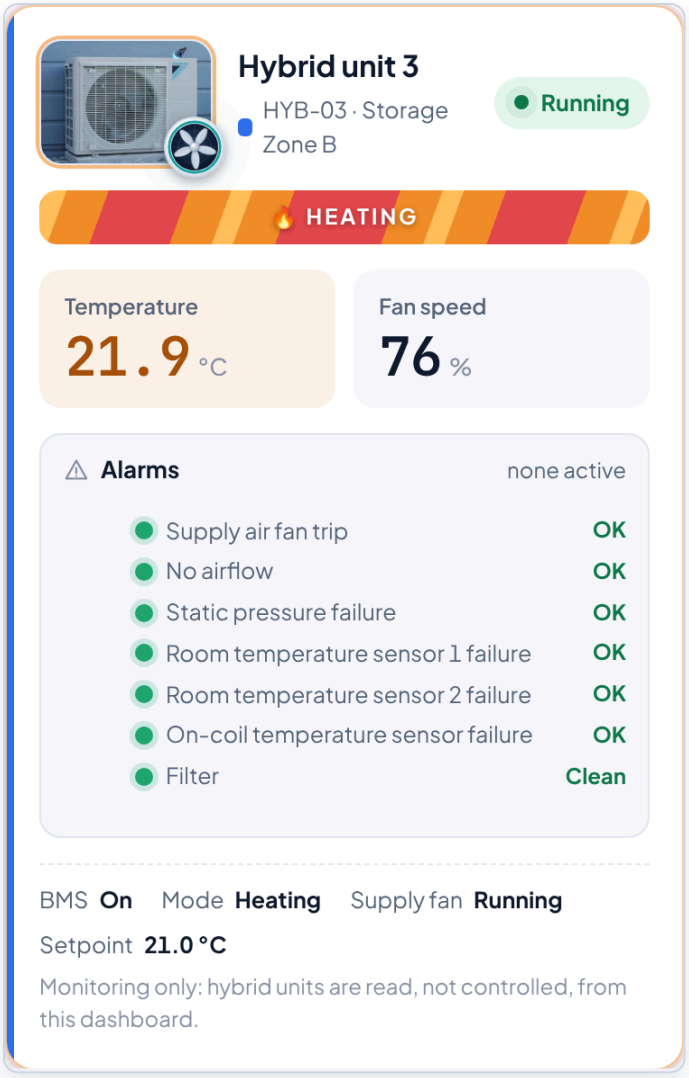
The room temperature changes slowly, over minutes to hours. It will not jump to the new setpoint. Watch the trend instead of changing the setpoint again.

Heating / cooling cannot be forced from the dashboard

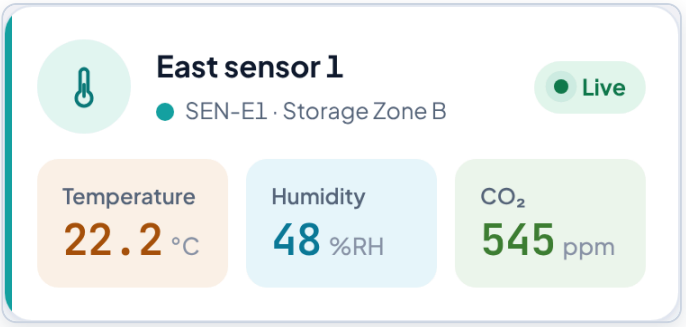
The mode is chosen by the unit itself and is shown for information only.

5 Hybrid units, room sensors & fire dampers

These are **monitoring only**: the dashboard reads them but never sends them commands.



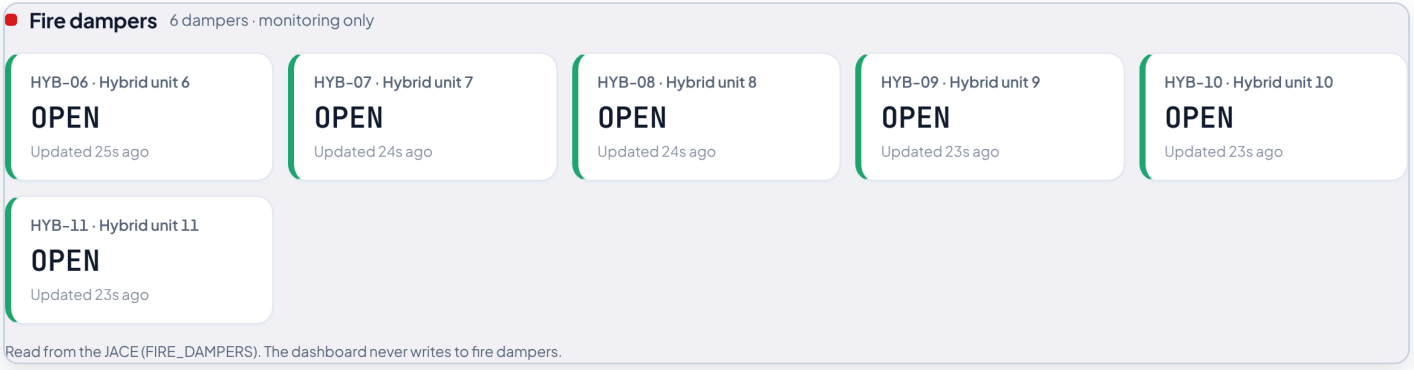
Hybrid unit: temperature, fan speed, BMS on/off, mode, supply fan, setpoint and its alarms.



Room sensor: temperature, humidity and CO_2 (ppm).

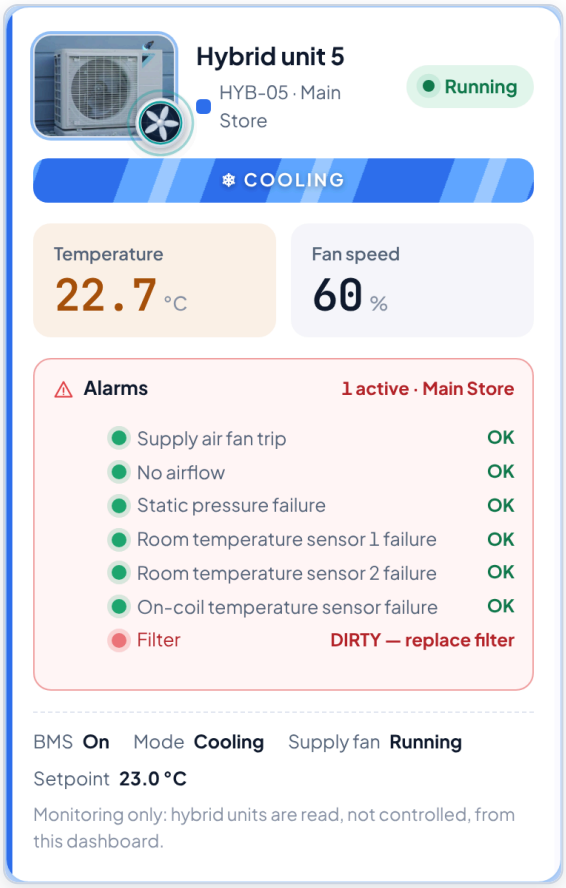
What you see

- **11 hybrid units**: read only. The amber band shows this unit is heating.
- **6 room sensors** (East and West): temperature, humidity and CO_2.
- **6 fire dampers** (hybrids 6–11): OPEN or CLOSED.



Fire dampers section at the bottom of the Units page.

6 Alarms



On each unit card

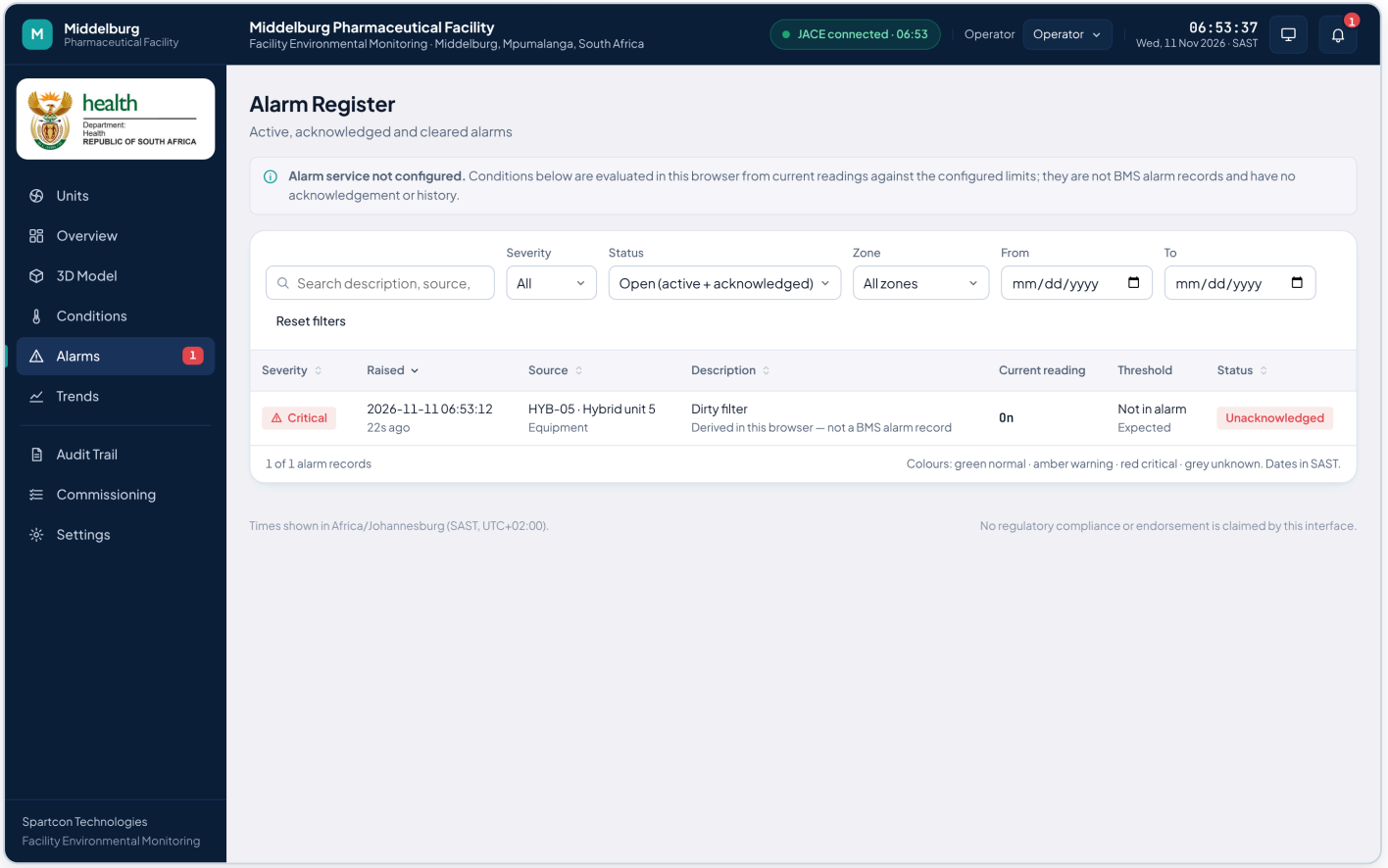
- **Green dot + OK**: normal.
- **Red blinking dot + ALARM**: the alarm is active. The panel header shows how many and the zone.
- **Filter**: green **Clean**, or red **DIRTY – replace filter**.
- **Grey NO DATA**: the alarm point cannot be read right now. This is not the same as OK.

Alarm points

- **Rooftop**: unit trip / alarm.
- **Hybrid**: supply air fan trip, no airflow, static pressure failure, room sensor 1/2 failure, on-coil sensor failure, dirty filter.

When an alarm appears

Note the unit, zone and time, check the unit if it is safe to do so, and report it to the IT administrator / maintenance. For a dirty filter, arrange a filter replacement.



The Alarms page lists every active alarm for the whole site. The red number on the bell (top right) and next to "Alarms" in the menu counts them.

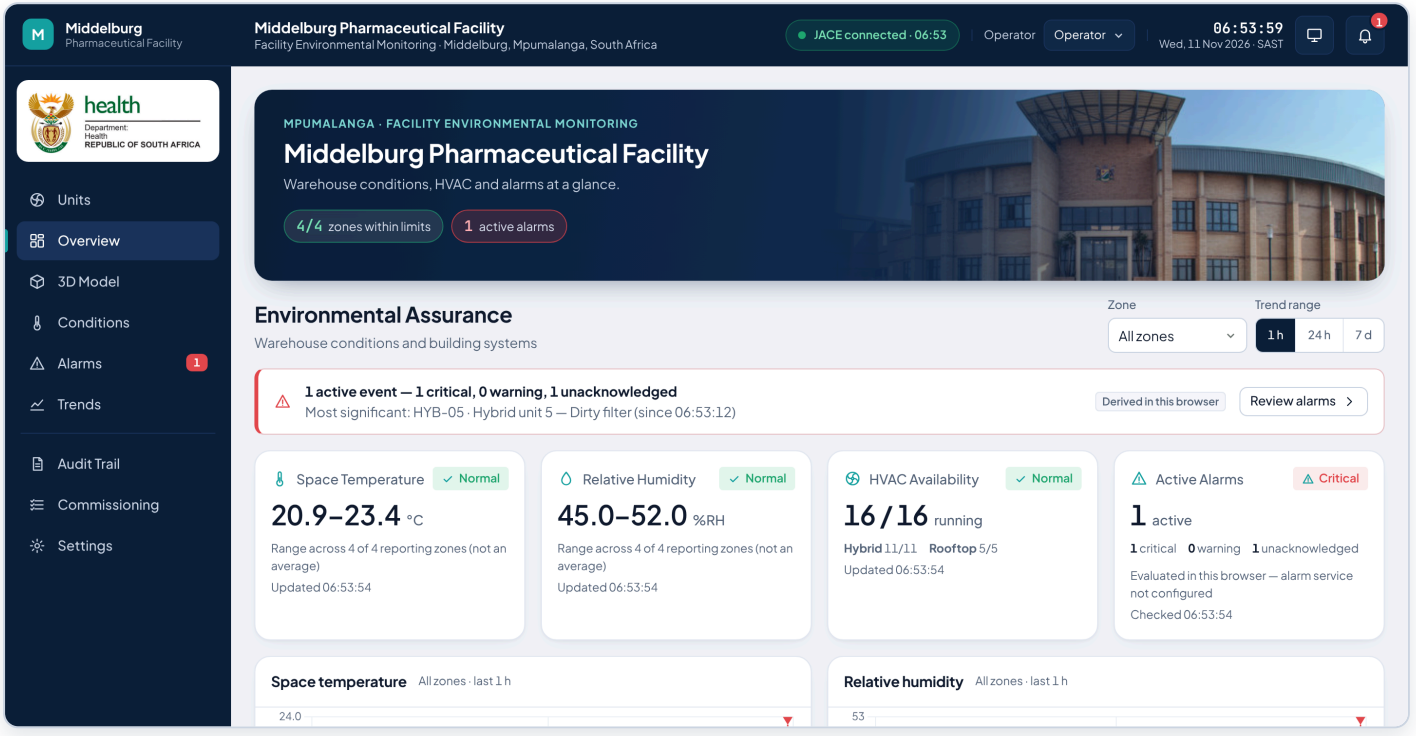
7 3D Model & Overview

3D Model: the warehouse at a glance



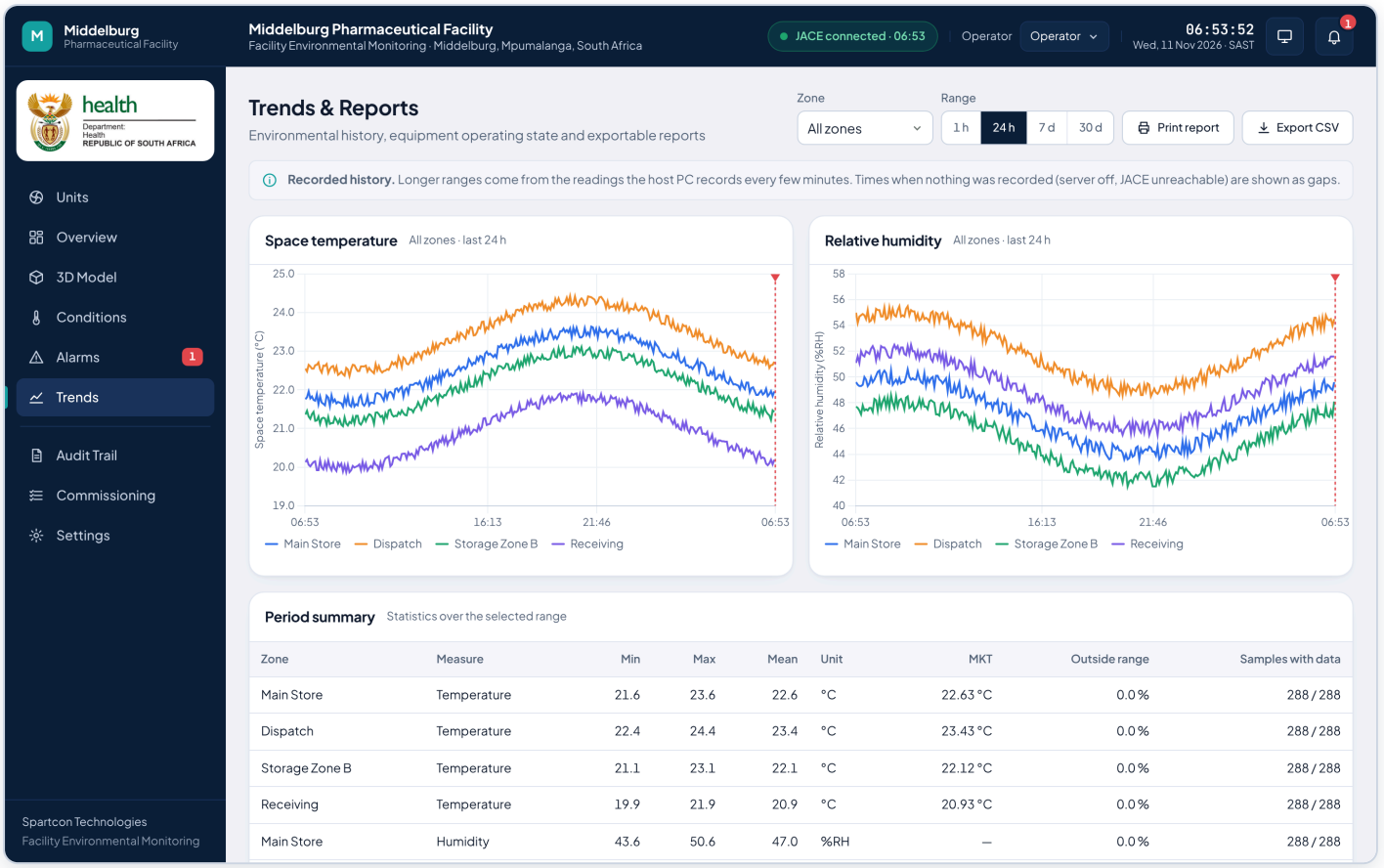
Zones are coloured by status (green normal, amber warning, red critical, grey no data). Rooftop units sit on the roof, hybrids on the floor, sensors as teal posts. Click a zone or a unit for details. "Plan" shows a top-down view.

Overview: site summary



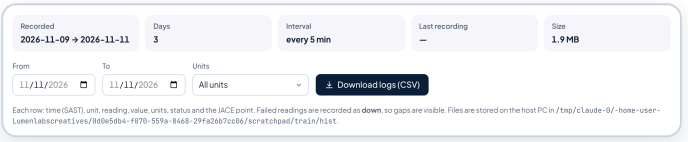
Key numbers, temperature and humidity of every zone, recent events and the facility location.

8 Trends & downloading logs



Trends: choose a zone and a range (1 h, 24 h, 7 d, 30 d). Below the charts: period summary (min / max / mean / MKT), excursions and unit run history. **Print report** makes a printable report.

Download the recorded data (Excel)



1. Go to **Trends** and scroll to **Stored logs**.
2. Choose **From** and **To** dates.
3. Choose **All units** or one unit.
4. Click **Download logs (CSV)** and open the file in Excel.

> Every reading of every unit and sensor is recorded every 5 minutes on the host PC, from the day the system went live, as long as the host PC is running.

9 Do's, don'ts & getting help

✔ Do

- Check the green **JACE connected** status before acting on readings.
- Send one command at a time and wait for **Confirmed**.
- Add a reason to commands; it is kept in the Audit Trail.
- Give setpoint changes time to take effect.
- Report red alarms and dirty filters promptly.
- Use the **Viewer** role on screens that only need to show data.

✘ Do not

- Open the **server cabinet**, or switch off, restart or unplug the **host PC**, router or JACE.
- Close the dashboard program on the host PC or change its settings.
- Use the **Settings** or **Commissioning** pages, or switch the data source to DEMO.
- Press Start/Stop repeatedly, or keep changing the setpoint while waiting for the room to respond.
- Share the remote-access (Tailscale) invite or your login with others.

What the labels mean

Label	Meaning
LIVE / Updated n s ago	Current value from the building controller.
— STALE	No new value recently (or connection lost). The small "last good" shows the last valid value and time.
— OFFLINE	The sensor or unit is not communicating.
— NOT MAPPED / NO DATA	No reading available for this point.
SIMULATED DATA banner	Demonstration mode, not real data. Contact the IT administrator.

Help and faults

For access, faults or questions, contact the IT administrator:

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